Mahmoud Yosri

ML Engineer

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PROFILE

Passionate and detail-oriented ML Engineer with a profound understanding of how machine learning works under the hood. A self-taught practitioner, I cultivated my mathematical and theoretical expertise through MIT OpenCourseWare (Linear Algebra) and rigorous independent study of foundational texts, including *Mathematics for Machine Learning*, *Practical Statistics*, and the *Deep Learning* textbooks by Ian Goodfellow and Christopher Bishop. I specialize in architecting neural networks from scratch and optimizing complex AI models, dedicated to bridging the gap between theoretical mathematics and practical, high-performance implementations.

EDUCATION

Bachelor of Computer Science Sep 2020 – Jun 2024
Kafrelshikh University, Faculty of Computer Science and Information Technology
GPA: 2.9/4.0
Relevant Coursework: Linear Algebra, Probability, Statistics, Data Structures, Algorithms, Software Engineering.

PROJECTS

Studio: Offline AI Background Removal & Photo Pipeline | Code | Case study *2026*

- Deployed two segmentation models (**IS-Net**, 1024 px, and **U^2-Net-P**, 320 px) with **ONNX Runtime** on the CPU: **1.2 s** and **0.2 s** per photo, fully offline.
- Wrote the post-processing in **NumPy**: mask cleanup, edge color decontamination (no halo), and white balance read only from neutral pixels so product colors stay true.
- Evaluated it on a known-truth bench of 8 real photos with injected defects: dark and blurry flagged **8 of 8**; one-tap white balance cuts color error (ΔE) from **10.6 to 3.5**; Fast vs Best agreement measured by IoU (0.99 median, 0.46 on busy scenes).

Argon: Job Matching and Market Analysis | Case study *2026*

- Scores each job ad from 0 to 100 against the user's CVs (**4,834 jobs**), with per-field skill weights learned from real ads and proposed changes approved by a person.
- Analysed **3,824 ads over 90 days** to rank in-demand skills and trends, and runs a local LLM (**Ollama**) so CVs never leave the machine.

Makemore Implementation (Autoregressive Language Model) *2025 – 2026*

- Developed a custom implementation of an autoregressive character-level language model from scratch, inspired by Andrej Karpathy's makemore library.
- Engineered diverse model architectures ranging from basic bigrams to complex Transformers akin to GPT, focusing heavily on under-the-hood mathematical operations.

Micrograd Implementation (Autograd Engine) *2025*

- Built a lightweight scalar library that implements core machine learning concepts, including backpropagation and neural networks, entirely from the ground up.
- Designed a PyTorch-like API, demonstrating a deep understanding of the mathematical foundations and calculus required to compute gradients efficiently.

AI Object Detection & Model Deployment *2024 – 2025*

- Deployed an SSD MobileNet model to enable real-time object detection capabilities.
- Integrated the ML model with a custom FastAPI backend, showcasing end-to-end model deployment and seamless API communication.

TECHNICAL SKILLS

- **Machine Learning & AI:** PyTorch, TensorFlow, ONNX Runtime, TFLite, Scikit-Learn, XGBoost, Transformers, Image Segmentation, Ollama.
- **Mathematics:** Linear Algebra, Probability, Statistics, Calculus for ML.
- **Programming Languages:** Python, C++, SQL, C#, Dart.
- **Data Science Libraries:** Pandas, NumPy, Pillow, Matplotlib, Seaborn.
- **Backend & Tools:** FastAPI, PostgreSQL, SQLite, pytest, Docker, Git, GitHub.
- **Spoken Languages:** Arabic (Native), English (B2).

CERTIFICATIONS

- **Deep Learning Specialization** | Coursera *2025 – 2026*
- **Mathematics for Machine Learning: Linear Algebra** | Coursera *2025 – 2026*
- **Machine Learning Specialization** | Coursera Verification *2024*
- **Python for Data Analysis** | National Telecommunication Institute (NTI) *2024*
- **HCIA-AI Learning Course** | Huawei ICT Academy *2023*